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RESEARCH

Cross-layer optimization in wireless sensor networks for energy-efficient multimedia transmission

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Abstract
Multimedia transmission (MMT) in wireless sensor networks (WSNs) presents unique challenges and considerations due to the resource-constrained nature of these networks. WSNs are typically composed of numerous small, low-power, and low-cost sensor nodes that collaborate to gather and transmit data from their surroundings, where optimizing communication across these heterogeneous nodes can be complex. To overcome this issue this paper proposes a cross-layer approach for MMT in WSNs through cross-layer optimization for cluster head (CH) selection and optimal routing of multimedia data without delay. The proposed methodology integrates a unified optimization model known as the Prey- predator Chasing (PPC) optimizerto enhance network clustering methods, and network path selection offering robust solutions for the efficient transmission of multimedia data within WSNs. The synergistic integration of these hybrid optimization algorithms harnesses their collective strengths, enabling improved resource allocation, data routing, and network performance. By employing PPC, this research advances the capabilities of WSNs in handling multimedia traffic, contributing to the development of more reliable and efficient WSNswith both energy efficiency and minimum data transmission delay. The results of the research show that node power utilization is reduced by implementing a cross-layer optimization method, and the throughput is also increased.

Keywords Wireless sensor network · Cross-layer optimization · Preypredator chasing optimization · Network clustering · Optimal data routing

1 Introduction

Due to the constrained resources and energy of the individual sensor nodes, MMTin a WSN poses special difficulties. WSNs are intended to collect and transmit data from the surroundings, typically for uses like surveillance, healthcare, and environmental monitoring [1, 2]. Images, audio, and video are examples of multimedia data that are data-intensive and require careful consideration to ensure effective transmission and energy conservation. Because multimedia data can be quite large, it is essential to use effective compression algorithms. WSNs can be configured to use well-known compression formats like JPEG for images and MPEG for video [3]. The quality and delay needs for various multimedia applications vary. To save energy or increase transmission reliability, it might be acceptable in some circumstances to slightly lower the quality. It's crucial to strike a balance between QoS demands and energy restrictions. The data transmitted can be significantly decreased by aggregating data from multiple sensor nodes before transmission [4]. Only pertinent information or condensed data is sent, rather than the raw multimedia from each node [5, 6]. WSNs have multiple layers of communication, including physical, MAC (Medium Access Control), and network. Combining these layers' optimizations can improve transmission effectiveness. To acquire and transfer multimedia data, nodes can awaken in turn. To make sure that nodes with multimedia data to transmit are alert when required, careful scheduling is required [7].

MMT may not be a good fit for conventional routing protocols. To guarantee that multimedia data is delivered with a

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minimum amount of delay and acceptable quality, QoS-aware routing protocols should be used [7, 8]. Wireless channels are noisy, so mistakes can happen during transmission. It's critical to implement mechanisms for error detection and correction as well as retransmission techniques to guarantee the integrity of multimedia data. Processing that takes place in-network can cut down on the amount of information that must be sent. Before transmitting pertinent data, some processing tasks, such as feature extraction from image data or stereo, can be carried out locally on the sensor nodes. MMT can be effectively accomplished by combining various sensor nodes with different capabilities. To ensure that crucial multimedia content is transmitted with higher priority and lower the chance of losing important data, multimedia data can be prioritized according to its importance [9]. Data quality, energy conservation, and network efficiency must all be carefully balanced when sending multimedia data over a WSN [10, 11]. The requirements of the application and the properties of the sensor network itself will determine the precise strategies that are used. Lossless MMT in a WSN is the practice of sending multimedia data (like images, audio, or video) over a network without suffering any data loss. This is crucial in applications like medical image analysis, monitoring systems, and empirical studies data where data accuracy and integrity are essential [12, 13].

In a WSN, where resources are limited, achieving lossless transmission necessitates careful consideration of several factors. While conventional lossy compression techniques, such as JPEG and MPEG, are frequently used for multimedia data, lossless compression algorithms should be taken into account for applications requiring accurate data reproduction [14, 15]. WSNs can be configured to use lossless codecs for audio and video and PNG for images. Even though lossless encoding might not be able to achieve as high ratios as lossy compression, it guarantees that no data is lost during decompression. Identification and correction of transmission errors can be aided by putting error detection and correction mechanisms in place at the physical layer. Implement trustworthy communication protocols that ensure the delivery of data packets to their intended location [16]. Error checking and retransmission of lost packets are features of protocols like TCP [17] (Transmission Control Protocol). Lossless data can be given priority over less important traffic by QoS-aware routing algorithms [18]. The communication protocol should include acknowledgment mechanisms. Create the WSN topology to reduce signal interference, boost signal power, and improve network performance as a whole. This may result in data transmission that is more dependable and lossless. Data compression, error detection and correction, dependable communication protocols, and careful network design are all necessary for uncompressed MMT in a wireless sensor network [19]. Even though several research has been implemented to enhance the transmission delay in WSNs. But in some approaches, multimedia data packet loss is upsurged due to the intervention of scalar data and control messages in WSNs [20]. Moreover, they suffered from the computational overhead, path establishment and slot scheduling [21]. Before all the data packets reach the destination the overflow of sensor buffer is occurred in early due to the larger size of the multimedia data. Thus, the higher priority multimedia data is discarded [22].

This research represents a vital contribution to the ongoing evolution of WSNs by significantly enhancing their capabilities in managing multimedia traffic. By optimizing energy efficiency and minimizing data transmission delays, these advancements promise to make WSNs more reliable and efficient. The main contribution of this research is as follows;

- **Cross-layer multimedia data transmission (CLMDT):** The integration of a cross-layer optimization approach across the WSN sensor node layers like the Network layer, MAC layer, and Physical layers achieves a balance between optimizing energy consumption and preserving data quality. By employing cross-layer parameter optimization, it becomes possible to reduce transmission delay and elevate the overall Quality of Service (QoS) for every node in the network. The node with the maximum optimal index in the cluster is selected as CH, and within a WSN the fitness evaluation for path selection is used for data routing.
- **Prey-predator Chasing (PPC) optimizer:** The PPC algorithm is designed to solve optimization problems, and it mimics the behavior of search agents hunting for food in a dynamic environment, which helps in optimizing cross layers of the sensor node in a WSN network and helps to find the optimal data routing path. The PPC optimization algorithm minimizes the computational time, improves the convergence accuracy with less amount of iterations, these advantages lead to the effective and robust.

The remaining of the paper is arranged as follows, Sect. 2 contains the literature review of the existing research, sedition 3 states the problem statement, Sect. 4 contains the WSN system model, Sect. 5 gives the explanation of network clustering and PPC optimization method, in Sect. 6 the result and analysis of the PPC-CLMDT system is discussed and portrayed, and Sect. 7 end the research with the conclusion.

2 Literature evaluation

This section categorized the existing solutions into routing based approaches, and cluster head selection or data transmission based approaches.

2.1 Routing based approaches

A brand-new Density-Based Clustering (DBC) method is presented by G. Kadiravan et al. [1] to improve the energy efficiency of WMSN. To solve hot-spot issues and increase network lifetime, the DBC technique's performance can be further improved by merging with bio-inspired algorithms. Prakash, V. et al. [23] suggested the modified particle swarm optimization (M-PSO) approach and Genetic algorithm (GA) to choose the non-cluster heads and Cluster Heads (CH). The suggested M-PSO algorithm calculated the probability of selecting the optimal nodes as the CH, moreover the GA was applied to determine the best shortest route and the best path was selected depend on the applied objection function.Singh, S.,et al. [24] developed a GA inspired greedy strategy based mutation process for the IoT based different WSNs distributed structure in smart city. Moreover, the suggested protocol designed a weighted fitness function and the issues of deployment process was overcomed to locate the nodes in effective way.

2.2 Cluster head selection or data transmission based approaches

To define non-correlated, and node-disjoint paths to transfer media data from source materials to the receiver, Imen Jemili et al. [6] developed a cross-layer multi-path routing approach. When choosing the best next-hop from among the eligible predecessors, additional contextual information, such as residual energy and implications for alternative routes, may be taken into account. A fusion-based WMSN framework was created by Adnan Yazici, et al. [7] to minimize transmitted over the internet amount of data that must be by intra-node processing. It demonstrates how multimodal data gathered with multimedia sensors that can record, store, and transmit multimedia data can be helpful. One of the issues with WMSNs is the transmission of large amounts of data to the CH. Another significant issue for the WMSNs is the effective transmission of data by the WMS nodes to the CHdue to the WMS nodes' limited energy. The virtual machine infrastructure for application predictive QoS pro-visioning more than 5G multimedia big data wifi networks was created by Xi Zhang et al. [9] which helps to efficiently balance the available energy. Jianbo Du et al. [3] suggest a method to jointly optimize the viewport rendering offloading and downlink transmit power control to reduce the long-term energy usage of wireless access-based MEC systems for elevated interactive Virtual video services support. The effectiveness of video streaming applications for dependable CoAP communications is evaluated by Waqas Ur Rahman et al. [10]. For dependable communications between IoT devices, the Constrained Application Protocol (CoAP) offers a congestion control mechanism. Some of the default settings for the congestion control mechanism were a result of the wireless sensor network's ability to have delayed responses. Applications that stream video, however, are delay-sensitive. However, considering CoAP transmission parameters along with the throughput and buffer of streaming over CoAP can increase the effectiveness of the quality adaptation algorithms. For safe data transmission over a network, Siddharth Singh and Raaghavv Devgon [5] established an assessment of encryption and lossless compression methods with additional safety. The JPEG XR encoding technique and block scrambling for image cryptography were the most effective standards used. Singh, S. and Malik, A. [25] suggested a Sink Mobility based on CH Energy (SMCHE) approach to manage and control the motion of the sinks. The selection of CH in Improved Dual-Hop Routing protocol (IDHR) and Multiple data sink-based Energy-Efficient Cluster-based routing protocol (MEEC) was achieved through the parameters of node density and other parameters like distance and energy among the sink and the node. Singh, S.et al. [26] suggested a GA-based active transmission of information for the transmittable diseases in the environment of IoMT. The various sinks and direct data assortment concepts are developed in the suggested approach to enhance the throughput and minimize the motion of the data packets.

2.3 Challenges

➢ Low latency is necessary for real-time multimedia applications, but WSNs may incorporate significant delays to routing, aggregation of data, and resource constraints, making it difficult to maintain low latency while preserving data integrity.

➢ Signal intervention, noise, and packet loss are common in wireless communication, which can lead to corrupted multimedia data. It can be difficult to guarantee lossless or acceptable loss transmission, especially for delicate applications.

➢ Scalability problems may result from the deployment of sensor nodes on a large scale. The network may have trouble efficiently handling multimedia transmission, routing, and aggregation as the number of nodes rises.

➢ WSN deployments take place in dynamic environments where network conditions are subject to quick changes. It is a difficult task to adjust the QoS parameters, compression algorithms, and transmission strategies to changing circumstances.

➢ WSNs may involve a variety of multimedia data types, including audio, video, and images. The network design is made more complex by the need for unique compression methods and transmission strategies for each type.

➢ Wireless channels may experience interference, which weakens and reduces the signal's strength. It becomes

challenging to keep up trustworthy communication channels, especially in demanding environments.

2.3.1 Motivation

MMTin WSNs are motivated by the potential benefits and wide-ranging applications that can significantly enhance various domains. Simple scalar sensor readings do not offer a complete picture of the environment's multimedia data, such as images, audio, and video. For applications, like environmental monitoring, wildlife tracking, and disaster response, these richer data may provide insightful information. Real-time situational awareness can be provided by multimedia data in applications, like security and surveillance monitoring. Video feeds are more effective than conventional sensor data at identifying anomalies, intrusions, and potential threats. Multimedia data transmitted by WSNs can act as remote sensors in remote or inaccessible locations, allowing scientists and researchers to gather data on any sensing area from the remote location. WSNs with multimedia data can support interactive applications, like augmented reality, virtual reality, and immersive experiences, and increasing user interaction. The ability of WSNs to collect rich, visual, and auditory data that can revolutionize a variety of industries and applications drivesmultimedia transmission, which allows individuals to make informed decisions, advance scientific knowledge, and develop novel experiences.

3 Problem statement

In WSN, the network is typically organized into different layers, with each layer holding its own set of protocols and functions. Cross-layer communication in WSN takes into account, the information and requirements from multiple layers simultaneously when making intellectual decisions. The primary objective of WSNs is to efficiently collect and transmit the data using the sensor nodes to the destination while conserving the energy resources of the sensor nodes, which is the fundamental trade-off in WSN design because more aggressive data transmission can lead to higher energy consumption. The CH is a key node within a cluster of sensor nodes and plays a crucial role in managing intra-cluster communication and these CH nodes are responsible for forwarding the data from source nodes to the BS. The choice of data forwarder nodes is equally important as CH selection because they contribute to the overall routing and data transmission. In a WSN, energy efficiency is a critical concern as the sensor nodes are powered by batteries and have restricted energy. Thereby, prolonging the networks' lifetime by conserving the energy of the individual nodes. Therefore, the research objective emphasizes the need to consider the requirements and constraints from multiple layers of the WSN protocol stack, which means taking into account factors from the WSN layers when selecting CHs and data forwarder nodes. By considering all these cross-layer requirements, the objective is to strike a balance that optimizes both minimum data transmission delay and energy efficiency, where the overall fitness of the ith sensor node in a WSN network is given by;

$$O_{ij} = Max[F(Ni)] \tag{1}$$

$$F(Ni) = \max\left(E_{ij}, Q_i, T_{ij}\right) \tag{2}$$

where E is the energy of the sensor node, Q is the quality factor of the sensor node regarding transmission delay, throughput, and traffic rate, and T is the direct and indirect trust of the node.

4 WSN system model

A WSN is designed to serve as a distributed and interconnected infrastructure for transmitting data in various applications, such as surveillance, environmental monitoring, and healthcare. In this system model, sensor nodes capture and process data locally, and then, cooperatively relay this data to a central CHor BS for further processing. The challenge in such networks lies in efficiently managing the high data rate, latency-sensitive nature of multimedia, and the resource-constrained nature of sensor nodes while optimizing energy consumption and ensuring reliable data delivery.Cross-layer optimization is essential for MMTin WSNs due to the unique requirements and challenges posed by multimedia data. These challenges include the need for efficient resource allocation to ensure timely and reliable delivery of multimedia content, issues related to energy conservation, and the transmission of multimedia data can be particularly power-intensive. Furthermore, mitigating the impact of interference and maintaining network stability are key challenges in ensuring the successful transmission of multimedia content in WSNs, making cross-layer optimization an essential approach for addressing these complex issues. Cross-layer optimization allows for the efficient coordination of multiple layers of the communication protocol stack to meet these multimedia-specific requirements, which can dynamically allocate network resources, such as bandwidth, power, and routing paths, to prioritize multimedia data streams that ensure the multimedia packets are transmitted with low latency, minimal jitter, and minimal packet loss, meeting the stringent QoS demands of multimedia applications. Furthermore, Cross-layer optimization enables intelligent adaptation to changing network conditions, such as varying link quality or congestion. Additionally, MMTin WSNs often involve data compression, error correction coding, and energy-efficient data sensing and processing. Cross-layer

approaches allow for the seamless integration of these techniques across multiple layers, optimizing energy consumption while preserving the data quality. Figure 1 represents the WSN system model.

4.1 Energy modeling of WSN

In wireless communication, the characteristics of the wireless channels are such that they are flexible and allow bidirectional communication between both the sender and receiver. This bidirectional communication is essential for the exchange of information between devices in the network. When considering the energy used by the transmitter in the network model, it becomes apparent that this energy expenditure is a crucial aspect of the system's operation. The transmitter's energy consumption is a critical factor that influences the overall performance and sustainability of the wireless network. It is essential to carefully manage and optimize the energy usage of the transmitter to ensure efficient and reliable communication within this bidirectional, wireless network environment [27]. The energy consumed during data transmission is denoted by,

$$\Xi_T(i,j) = \Xi_{T-elec} \times i + \mathrm{E}_F \times i \times D^2, D < D_0 \tag{3}$$

$$\Xi_T(i,j) = \Xi_{T-elec} \times i + \mathrm{E}_M \times i \times D^4, D \geq D_0 \tag{4}$$

where, the energy expended during the transmission of a bit of data, denoted as Ξ_{T-elec} that is influenced by factors E_F and E_M related to the transmission energy consumption model. E_F is free-space transmission, while E_M is multipath loss. The distinguishing factor between the two models lies in the boundary condition expressed as D_0. Specifically, when the transmission distance, D, surpasses the threshold D_0, there is an observable increase in the transmitter's energy consumption, which can be attributed to the fact that as the transmission distance grows beyond D_0, the signal propagation and amplification require more energy. On the receiving end, the energy utilization for receiving a bit of data can be described using the following equation (as expressed in Eq. 5). This equation quantifies the energy expenditure at the receiver,

$$\Xi_R(i,j) = \Xi_{T-elec} \times i \tag{5}$$

4.2 Delay analysis

Data transmission delay is the time taken for data to traverse the network from source nodes to the BS. Minimizing delay is important in applications where real-time data is crucial. However, reducing delay often comes at the cost of increased energy consumption due to more frequent data transmission. The delay analysis in WSNs involves the evaluation of various factors that contribute to the delays within the network and analyzing the delays in WSNs is essential for designing efficient communication protocols, optimizing network performance, and meeting application-specific requirements.The total delay can be represented mathematically as follows:

$$D_T = D_{Tx} + D_{ptop} + D_{\mathrm{Proc}} + D_{que} + D_{wait} \tag{6}$$

where the total delay is calculated by combining transmission delay D_{Tx}, propagation delay D_{ptop}, processing delay

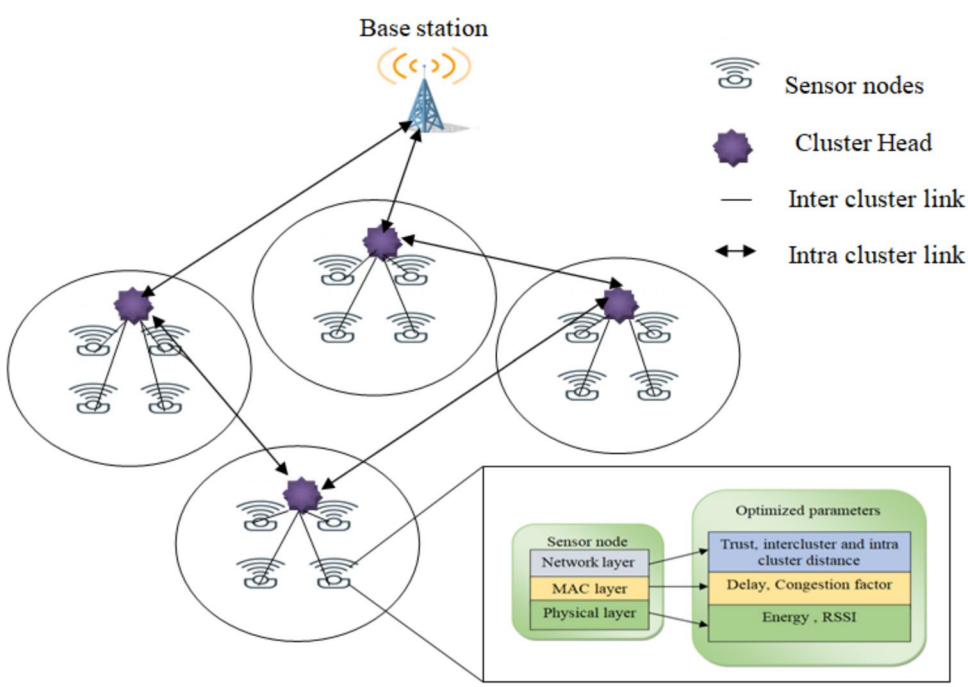


Fig. 1 Cross-layer optimization in wireless sensor network nodes

D_{Proc}, queuing delay D_{que}, and waiting or synchronization delays D_{wait}.

4.3 Throughput analysis

Throughput analysis typically involves calculating the rate at which the data packets are successfully transmitted from sensor nodes to the destination. The throughput considers the factors, such as network topology, routing protocols, data packet size, and communication overhead. A higher throughput indicates a more efficient use of the network resources, which is essential for applications demanding timely data delivery. The accurate throughput analysis helps in optimizing the network design, choosing suitable routing strategies, and ensuring that the WSN can meet the data transfer requirements of multimedia transmission. The mathematical equation for throughput in a WSN can be expressed as:

$$T = \left\{ \left(\frac{S}{N * P} \right) * \left[1 - e^{\wedge}\left(-\lambda * T_{packet} \right) \right] \right\} \tag{7}$$

where T is the throughput, S is the packet size, and N is the number of nodes. P is the packet error rate (PER), representing the probability that a packet is received correctly, λ is the packet generation rate, and T_{packet} is the packet transmission time (in seconds), which is calculated as;

$$T_{packet} = \frac{S}{R} \tag{8}$$

where R is the data rate of the communication link (in bits per second).This equation takes into account the packet size, the probability of successful packet transmission (accounting for possible errors), and the packet generation rate, which provides a measure of the network data transfer capacity, considering factors that affect the successful delivery of the data in WSN.

4.4 Traffic rate control

Traffic rate control in WSNs typically involves managing the data rate at which sensor nodes transmit information to the CHor data collection point. It is a critical aspect of network management and optimization and involves mechanisms and strategies to regulate the rate at which data is generated, transmitted, and processed within the network. The primary goal of traffic rate control is to ensure efficient utilization of the limited network resources, including energy, bandwidth, and processing capacity while meeting application-specific requirements.The coordination between layers, such as network and data link layers allows for intelligent decisions regarding frame sizes, duty cycling, and route selection to achieve efficient traffic rate control.

4.5 Trust degree of WSN nodes

The trust degree of the nodes involves assessing the reliability and trustworthiness of each node based on various factors and metrics. Trust management in WSNs is critical for ensuring the integrity and security of data in a network, where sensor nodes may be susceptible to various attacks and failures.The trust score for a node can be calculated as follows:

$$Trust_{node} = \left\{ \left(W_1 * N_1 \right) + \left(W_2 * N_2 \right) + \left(W_3 * N_3 \right) + \cdots \left(W_n * N_n \right) \right\} \tag{9}$$

where W are the weights assigned to the trust metrics, and N are the normalized values of the trust metrics. A high trust degree implies confidence in the integrity and authenticity of the information exchanged between nodes, ensuring that data is not compromised or tampered with. By incorporating trust-awareness into cross-layer design, WSNs can enhance network performance, security, and overall reliability, ultimately leading to more robust and efficient sensor network deployments.

a) Direct Trust Model: Direct trust refers to the trustworthiness of a node based on its past interactions and behavior. Let's denote direct trust as $DT(i)$ for node i as

$$DT(i) = [\alpha * ST(i)] + [\beta * R(i)] + [\gamma * SB(i)] \tag{10}$$

where α, β, γ are weight coefficients that determine the importance of each factor, $ST(i)$ represent the number or percentage of successful interactions involving the node i ,$R(i)$ assess how often the node i is available and responsive, and $SB(i)$ measure the node's adherence to security protocols and absence of malicious behavior.

b) Indirect Trust Model: Indirect trust considers trust recommendations from other nodes in the network often based on their direct trust values. It can be calculated as an aggregation of trust values received from neighboring nodes. Let's denote indirect trust as $IDT(i)$ for node i as

$$IDT(i) = \sum DT(i) * W_{ij} \tag{11}$$

where $IDT(i)$ is the indirect trust of the node i, $DT(i)$ is the direct trust of the node j, and W_{ij} represents the weight assigned to the trust recommendation from node j to node i. All the above-mentioned parameters are important in multimedia transmission, so to improve the performance of the parameters, cross-layer optimization is done in the subsequent section.

5 Cross-layer optimization and factors optimized through the proposed algorithm in WSN

Cross-layer optimization facilitates the transmission of multimedia data within WSNs, which arises from the unique demands and complexities associated with multimedia data. Cross-layer optimization allows for the efficient coordination and collaboration among various layers within the network's communication protocol. Moreover, the Cross-layer approach across the network, MAC, and Physical layers achieves a balance between optimizing energy consumption and preserving data quality. Each layer shares the details across different layers based on several metrics like distance, trust, delay, energy, congestion factor and RSSI. Accordingly, the network layer uses the metrics like trust, inter cluster and intra cluster distance in order to make communication with other layers. Moreover, the MAC layer considers delay, and convergence factor to share the details, whereas the physical layer uses RSSI and energy factor for sharing the details with other layers. The evaluation of the optimal index inside a node is shown in Fig. 2.

5.1 Cross-layer parameter optimization

In the following subsection, a detailed explanation of the cross-layer parameter optimization involved in calculating the optimal indexfor individual sensor nodes in WSN [28, 29]. When it comes to managing and optimizing the communication within a WSN, the determination of cross- the optimal indexplays a crucial role, which is essential for making informed decisions regarding the selection of CH, and data transmission. The layer parameters considered for the cross-layer optimization are detailed below.

a) Network layer parameters

i) Intra-Cluster Distance:

Intra-Cluster Distance (Intra-CD) refers to the distance between sensor nodes within the same cluster, which is used to assess the compactness of nodes within a cluster. A smaller Intra-Cluster Distance indicates that nodes are tightly clustered, while a larger distance suggests that nodes are more spreadout, which is mathematically modeled as,

$$Intra - CD = \frac{1}{N} \sum \left[d(node_i, center_cluster1)\hat{}2 \right] \quad (12)$$

In Eq. (12), the mentioned factors assess the degree of node dispersion within a cluster.

ii) Inter-Cluster Distance

Inter-cluster distance (ICD) refers to the separation or spacing among individual clusters within a WSN.ICD refers to the distance between the centers of different clusters in a WSN that is the crucial parameter for cluster-based routing protocols. ICD is mathematically modeled as,

$$ICD = d(center_cluster1, center_cluster2) \quad (13)$$

where,ICD is the Inter-Cluster Distance, d is a distance metric,$center_cluster$ are the coordinates of the centers of the cluster's distance being compared.

iii) Trust

Trust is used to evaluate and quantify the trustworthiness of nodes within the network. These models are essential for addressing security and reliability concerns in WSNs, where nodes often operate in unattended or hostile environments.

b) Medium Access Control (MAC) layer parameters

The MAC layer is a reliable and efficient communication link between WSN nodes. The MAC protocol comprises a set of rules governing the transmission of data by individual nodes over the commonly shared wireless medium. Following are the MAC layer parameters to be optimized.

i) Transmission Delay: MAC layer delays can collectively contribute to the overall latency and performance of a network thereby, minimizing and managing these delays is essential for ensuring efficient and reliable cross-layer communication.The initial access delay is denoted by D_0 the delay of Rth retransmission is denoted by D_R. The total access delay in case of R retransmissions in a sensor node is,

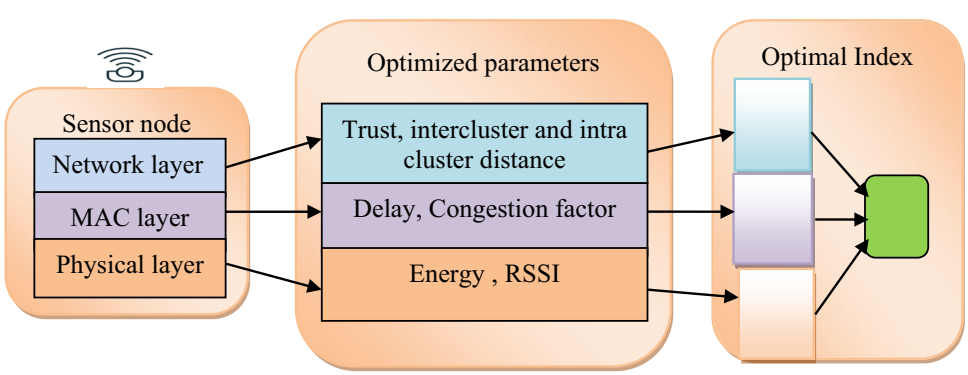


Fig. 2 Optimal index evaluation inside a node

$$D = \sum_{i=0}^{R} D_i \tag{14}$$

ii) Congestion factor: The MAC layer is responsible for managing access to the shared communication medium. Congestion in the MAC layer can significantly impact network performance by causing delays, packet collisions, and reduced throughput.A mathematical model for the congestion factor in the MAC layer is denoted as,

$$CF = \left(\frac{\lambda * T}{N * C}\right) \tag{15}$$

where, λ is the arrival rate of new packets per second, T is the MAC layer frame duration, and C is the MAC layer channel capacity.The congestion factor in the MAC layer for cross-layer communication plays a crucial role in managing network congestion and optimizing communication in wireless networks, which is represented as,

$$CF = \left(\frac{P_busy}{P_total}\right) \tag{16}$$

where, P_busy represents the portion of time when the channel is busy or occupied by ongoing transmissions. It can be calculated as the product of the average packet transmission time and the transmission rate. P_total represents the total time available in a given time interval for communication on the channel.

c) Physical Layer

The Physical Layer plays a critical role in facilitating communication among sensor nodes, which is essential to monitor and manage various parameters at this layer to ensure efficient network operation. Among the key parameters, two significant boundaries are meticulously tracked for each sensor node: leftover energy and Received Signal Strength Indicator (RSSI).

i) Energy: Leftover energy is a crucial metric as it directly influences a sensor node's role within the network. Nodes with higher remaining energy levels tend to be preferred for tasks, such as CH selection and routing decisions. This preference is driven by the need to prolong network lifetime and ensure reliable data transmission.Let $E_i(t)$ represent the leftover energy of the ith sensor node at a time t can be expressed as,

$$E_i(t) = E_i(0) - \sum_{j=1}^{k} P_{ij}(t) \cdot \delta t \tag{17}$$

where $E_i(0)$ is the initial energy of the ith node, $P_{ij}(t)$ represents the power consumption of the ith node for a specific operation or communication with the jth node at time t, and δt is the time interval.

ii) Received Signal Strength Indicator: RSSI estimation for each sensor node within the network is calculated separately. These RSSI values are then used to compute a threshold-based likelihood estimate for the RSSI boundary associated with each sensor The RSSI threshold at the current time t is determined by analyzing the reception rate of signal packets from the neighboring node, which provides a valuable measure of the network's signal quality and reception capabilities, aiding in efficient management and optimization.

$$RSSI_{ij}(t) = (RSSI_0) - 10 \cdot n \cdot \log_{10}\left(\frac{d_{ij}(t)}{d_0}\right) \tag{18}$$

where $(RSSI_0)$ is the RSSI value at a reference distance d from the transmitter, and n is the path loss exponent, which depends on the environment and frequency, and $d_{ij}(t)$ is the distance between the ith and j^{th} nodes at the time t.

5.2 Evaluation of optimal index

The cross-layer evaluation of the optimal index involves a holistic assessment of sensor nodes' capabilities and attributes across the network, physical, and MAC layers. The resulting cross-layer likelihood value plays a crucial role in selecting sensor nodes for CH within the cluster network, ultimately enhancing its overall performance and efficiency. The optimal index of these layers is calculated using,

$$I = \alpha_1 P_N + \alpha_2 P_M + \alpha_3 P_P \tag{19}$$

where P_N is the network layer parameter, P_M is the MAC layer parameter, and P_p is the physical layer parameter. The value of $\alpha_1 + \alpha_2 + \alpha_2 = 1$, and is randomly decided. The node with the maximum optimal index in the cluster is selected as CH, in which these functions are optimally tuned by the proposed PPC optimization algorithm.

5.3 Network clustering

Once the optimal index is evaluated for all the nodes, the network clustering is performed based on the calculated index such that the lifetime and QoS of the network are enhanced. In network clustering, a PPCoptimization strategy drawing inspiration from nature is used for determining the CH nodes by taking into account the optimal index of the sensor nodes as presented in Eq. (19). The PPC optimization method serves as a robust foundation for network clustering methodology. These algorithms enable enhanced efficiency and accuracy in the selection of CH nodes within the WSN. This PPC optimization approach not only offers a comprehensive and efficient solution for CH selection but also adapts to the dynamic nature of WSNs, ensuring optimal performance under varying conditions. Through the utilization of PPC

optimization techniques, our methodology paves the way for more robust and intelligent network clustering in WSN. The cross-layer parameter optimization is used to decrease the transmission time, reduce delay, improve the QoS, and enhance optimal data routing.

5.4 Multimedia transmission using Prey-predator chasing optimization

Path selection is a critical component in the optimization of routing and path-finding algorithms within WSNs. It serves as the key for guiding the search for the most efficient and optimal path for data routing within the network. In WSNs, where energy conservation and reliable data transmission are paramount, the choice of the right path can have a profound impact on network performance. By determining the optimal route, path selection helps ensure that data is delivered accurately and efficiently, thereby extending the network's lifespan and enhancing its overall functionality. The optimized fitness evaluation for path selection in selecting the best network path in WSN for MMT is discussed in the following sections.

5.4.1 Network path

In a multimedia network, data transmission often involves a complex path from the source to the destination, typically comprising three key components: the sensor nodes, CH, and the BS. The journey begins at the sensor, where multimedia content is generated or sourced. From there, the data is relayed to a CH, which acts as a local coordinator within a cluster of interconnected devices. The CH optimizes data aggregation and routing decisions before forwarding the multimedia content to the BS. This network path ensures efficient data management and delivery in multimedia applications, balancing the distribution of processing tasks and optimizing bandwidth usage to maintain quality and reliability throughout the transmission.

5.4.2 Fitness evaluation for path selection

The fitness evaluation for path selection is a crucial aspect of optimizing routing and path-finding algorithms, which plays a fundamental role in guiding the search for optimal data routing. The path selection can be formulated as optimization problems and fitness functions are used to quantify the quality of paths based on their attributes. The fitness evaluation of the jth path is represented as,

$$F_j = a_1 \sum E_{ij} + a_2 \sum Q_{ij} + a_3 \sum T_{ij} \tag{20}$$

where $\sum E_{ij}$ is the energy of all the nodes in the jth path, Q_{ij} and is the QoS of transmission delay, throughput, network energy, and traffic rate.T_{ij} is the direct and indirect trust ofall the nodes in the jth path.

5.4.3 Prey-predator Chasing (PPC) optimization model

The PPC is modeled based on the hybrid characteristics of the Wolf Search Algorithm (WSA), Particle Swarm Optimization (PSO), and Cat Swarm Optimization (CSO). The PPC algorithm is designed to solve optimization problems, and it mimics the behavior of search agents hunting for food in a dynamic environment. When hunting for food, all the search agents keenly look for food, safely watching the environment regarding the availability of the location enemies and abundanceof food todecide the onset of searching, waiting, hunting, and safeguarding. Therefore, focusing on well-organized and decided hunting is interesting, and successful hunting depends on the experience, strength, and population of the search agents in the current search boundary. In this optimization, the entire population is organized under three groups based on the user-defined threshold, and k-means clustering.

a) Motivation

The PPC Optimization model is proposed based on the collective intelligence, and social behavior of PSO [30], the hunting strategies of WSA [31], and the CSO [32, 33] model for finding food. The social behavior of PSO is inspired by the collective behavior of birds and fish, which refers to how particles interact and share information as they search for the best solution. The hunting strategies of WSA draw inspiration from the coordinated and adaptive behavior exhibited by wolf packs in nature. Just as wolves work together to maximize their hunting success, WSA employs a population of agents that collaborate to search for optimal solutions in the solution space. These agents emulate the concept of alpha, beta, and omega wolves within a pack. By mimicking these hierarchical roles and the adaptive decision-making of wolf packs, WSA efficiently explores the solution space. CSO is motivated by the collective hunting behavior of cats in the animal kingdom. CSOs exhibit both individual and group behaviors. They explore the solution space independently, leveraging their knowledge, and periodically interact with others to exchange information about promising solutions. The PPC optimization model is selected due to its benefits in efficiency, because the process of merging and searching of independent solution is performed at the same time by every searching agent. Moreover, it is also selected due to its advantages, such as robustness, faster convergence, lower computational time, ability to resolve intricate issues and attainment of higher convergence accuracy with lesser number of iterations. As a result of these advantages, PPC optimization model minimizes the energy consumption and optimize the path of data routing, which leads to an more robust and effective multimedia transmission. So, it is well

suited for the cross-layer optimization in multimedia transmission compared to other optimization approaches.

5.4.4 Initialization

Initializing the population for the PPC algorithm involves creating a set of search agents that mimic the behavior of adaptive hunting. Generate the initial set of search agents, where each agent represents an individual in the population with characteristics such as Position in the solution space, experience, knowledge, and strength.Based on the characteristics of the N search agents, three groups using k-mean clustering are formed, which follows different and unique characters and update rule throughout the execution of the algorithm.

5.4.5 Fitness evaluation for solution setup

For the formation of the groups in the search space, k-means clustering setsthe cluster size as k = 3 and uses the fitness function described in Eq. (21). K-means clustering is used in sensor networks, to group sensor nodes into clusters based on the optimal index value, which helps to organize the sensor nodes for optimal routing of multimedia data transmission. The fitness function for forming the cluster is as follows,

$$F = \left(\alpha E_i + \beta S_i + \gamma A_i + \lambda M_i\right) \tag{21}$$

where E_i is the experience index, S_i is the strength index, A_i is the awareness index, and M_i is the maturity index. For optimal routing, the fitness value should be maximum. The experience index weight factor of the search agent is $\alpha \in (0,1)$, The fitness of the search agent is measured based on,

$$\beta = \frac{1}{N}\sum_{N}\left[F\left(P_i\right)^{t+1} - F\left(P_i\right)^{t}\right] \tag{22}$$

where $F\left(P_i\right)^{t+1}$ is the fitness of the search agent at the time $t+1$, The weight for the awareness index is,

$$\gamma = \omega_{\min}\frac{\omega_{\max} - \omega_{\min}}{F\left(P_i\right)^{t+1} - F\left(P_i\right)^{t}} \tag{23}$$

where ω represents the minimal and maximum value of the weight in the awareness index, the maturity index λ value is measured with,

$$\lambda = \omega_{\max}\frac{t_{\max}\left(\omega_{\max} - \omega_{\min}\right)}{t_{\max}} \tag{24}$$

The fitness value should be maximum upto the provided user preset threshold so that the clustering of the nodes is performed, where all the solutions are grouped under any of the three clusters based on their experience, strength, awareness, and maturity index.

5.4.6 Communication behaviour

After evaluating the fitness value three cases are used for position update, such as intelligent searchers, messengers, and scapegoats based on the fitness value as shown in Fig. 3. Among these groups, the intelligent searchers possess higher experience, strength, awareness, and maturity indices. They leverage their wealth of experience to seek out food sources and serve as vigilant sentinels, promptly alerting the group to potential threats. The messenger agents act as intermediaries, facilitating seamless communication between the searchers and scapegoats. Together, they gather vital information to safeguard the population and share valuable insights about food sources, enhancing the group's sustenance and enabling collective rest and group activities. The scapegoat, on the other hand, assumes a pivotal role in the entire operation, beyond the searchers and messengers. Their extensive exploration serves to shield the entire population of search agents from potential adversaries, ensuring the group's safety and security. The intelligent searchers' group is denoted as $F_{\max}$, the messenger agents group named as F_{med}, and the scapegoat agents group is named as $F_{\min}$.

5.4.7 Position update

Based on the fitness value the position is updated for intelligent searchers, messengers, and scapegoats, which is explained in the following sections,

i) case 1-if $F < 0.5$ **[Scapegoat]**

The scapegoat is involved in the entire activity of the search agents, which is assigned to explore the solution space more extensively, even beyond the regions explored by the searcher and messenger. The scapegoat helps protect the entire search agents in the population from enemies. The scapegoat in a population is represented as,

$$G^{t+1} = G^{t} + v^{t+1} \tag{25}$$

where G^{t+1} is the scapegoat agents in time $t+1$,G^{t} is the scapegoat agents in time t, and v^{t+1} is thevelocity of a search agent to locate food within a given search space at the time t, which is denoted as,

$$v^{t+1} = v^{t} + r_3 a_3\left(G_{per} - G^{t}\right) \tag{26}$$

where $r_1, r_2.r_3 \in (0,1)$ are random values, $a_1, a_2, and\ a_3$ are decreased from 2 to 0 over the course of iterations.

ii) case 2-if $0.5 > F \leq 0.9$ **[Messenger Agents]**

The messenger agents in the group communicate with the searcher and scapegoat and collectively gather information to protect the population of the group and share

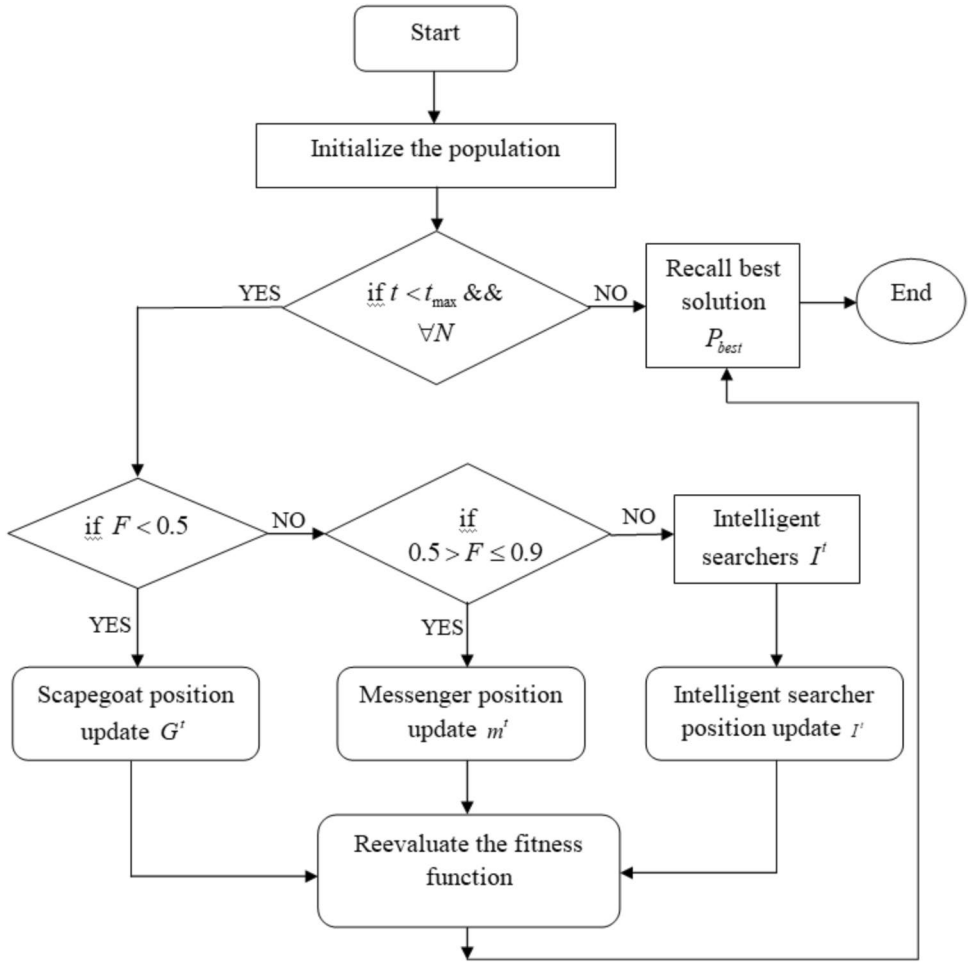


Fig. 3 Flow chart of Prey-predator Chasing Optimization-model

information regarding food to the population for feeding. The collective information helps the population for resting and any kind of group activities. The messengers' agents in the group are mathematically represented as,

$$m^{t+1} = m^t - R_2 \cdot D_2^t \tag{27}$$

where m represents the messenger in the group, R_2 is the messenger degree, and D_2^t is the distance between the messenger and the food in the search space, which is denoted as,

$$D_2^t = \left| m^{t+1} - W \right| \tag{28}$$

where W is the position of the food in the search space.

iii) case 3- if $F > 0.9$ [Intelligent searcher]

The intelligent searcher in the group has a higher experience index, strength index, awareness index, and maturity index, and they look for food based on their experience and alert the group based on the information about attackers. The intelligent searcher forages for food to provide to the entire pack, keep the pack nourished, and they also help the pack to move to a safer place. The intelligent search agents in time $t + 1$ are represented as,

$$I^{t+1} = I^t - R_1 \cdot D_1^t + rand * v^{t+1} \tag{29}$$

where I^{t+1} is the intelligent search agents in time $t + 1$,I^t is the intelligent search agents in time t, R_1 is the messenger degree,D_1^t is the distance between the intelligent searcher and the food, and v^{t+1} is thevelocity of a search agent to locate food within a given search space at time t.

Thevelocity of a search agent is given as,

$$v^{t+1} = v^t + r_1 a_1 \left(I_{best} - I^t \right) + r_2 a_2 \left(I_g - I^t \right) \tag{30}$$

where I_{best} is the personal best of the intelligent searcher, and I_g is the global best of the intelligent searcher. The distance between the intelligent searcher and the food is represented as,

$$D_1^t = \left| I^{t+1} - W \right| \tag{31}$$

where W is the position of the food in the search space.

5.4.8 Reevaluate the fitness function

After updating the position based on the fitness value reevaluate the fitness function incorporating clustering (19) and routing (20) equations, to optimize the network's performance. This holistic approach allows for the selection of network configurations that not only minimize energy consumption through clustering but also optimize data routing paths, ultimately leading to more robust and energy-efficient WSN designs. After evaluating the fitness function the best solution is declared for network clustering and data path routing. The flow chart of the proposed PPC is shown in Fig. 3. The pseudocode Prey-predator Chasing Optimizationmodel is shown in Table 1.

6 Result and analysis

The results and analysis for the Prey-Predator Chasing (PPC) Optimization-based cross-layer multimedia data transmission (PPC-CLMDT) are explained in the sections below.

7 Experimental tool

Using the Vision dataset, the PPC-CLMDT method is run on a Windows 10 with 16 GB of RAM using the Python software. The details about the simulation parameters are shown in Table 2

7.1 Performance metrics

The performance of the PPC-CLMDT is obtained by performance metrics such as throughput, alive node,

Table 2 Simulation parameter

Parameters	Values
Simulation Area	500 m x 500 m
No. of Nodes	200
No of Packets and Packet Size	100 packets/100 kb
Energy	1 J
Time	0.5 ms/packets
No. of Clusters	5

Table 1 Pseudocode for PPC model

S.No	Prey-predator Chasing (PPC) Optimization model
1.	Start
2.	Initialize the Population
3.	Evaluate the Fitness
4,	While $\left(t < t_{\max} \,\&\&\, \forall N\right)$
5.	do
6.	if$\left(F < 0.5\right)$
7.	{
8.	Spacegoat position update G^t
9.	}
10.	else if$\left(0.5 > F \le 0.9\right)$
11.	{
12.	Messenger Position update m^t
13.	}
14.	else
15.	{
16.	Intellegent Searcher Position Update I^t
17.	}
18.	Reevaluate the fitness function
19.	Declare the best solution P_{best}
20.	Terminate the process
21.	End While

and average energy left. The performance is analyzed by changing the number of nodes in the WSN, which is 50, 100, 150, and 200 nodes for 2000 rounds.

a) Throughput

Throughput analysis in WSN involves estimating the amount of data that is successfully transmitted through the sensor nodes over a certain period. The mathematical formula of throughput is:

$$T = \frac{totaldatasent}{totaltime} \tag{32}$$

The total data sent depends on the data rate per node and the number of nodes. The total time includes the time taken for data transmission and communication overhead.

b) Alive node

The alive node is calculated based on the number of nodes that are alive after a successive number of rounds. The degradation in performance is due to the draining of battery charge in the sensor node.

c) Average Energy Left

The average energy left in the WSN is measured based on the remaining energy present in the sensor nodes with an increase in the number of rounds.

7.2 Simulation results

Figure 4(a) and 4(b) shows the simulation results of PPC-CLMDT. It shows the visual formations of each cluster and the group of nodes with blue, red, green, yellow, blacks colours represent the different clusters and the cluster heads are selected based on the fitness measure and the fitness is evaluated by the parameters, such as energy, delay and trust.

7.3 Comparative analysis

The comparative analysis of the PPC-CLMDT method's performance was done by comparing PPC-CLMDT with existing methods likeCL-CFMPR: Cross-Layer Correlation-Free MultiPath Routing [6], WSN- Nature-Inspired algorithm-based Cross-layer Clustering (WSN-NICC) protocol [28], SMCHE [25], M-PSO [23], CSO-CLMDT [32, 33], PSO-CLMDT [30], and WSO-CLMDT [31] models.

a) Comparative Analysis for 50 nodes

The comparative analysis of PPC-CLMDT and existing approaches with 50 nodes is shown in Fig. 5, where the total number of alive nodes for 1000 rounds in the CL-CFMPR model is 9, and the total number of alive nodes for 1000 rounds in the WSN-NICC model is 17, SMCHE is 24, M-PSO is 13, CSO-CLMDT is 24, PSO-CLMDT is 25, and WSO-CLMDT is 26. Finally, the total number of alive nodes for 1000 rounds in the PPC-CLMDT model is 28. The average energy of 50 nodes of the PPC-CLMDT method for the 360 rounds is higher than the existing CL-CFMPR, WSN-NICC, SMCHE, M-PSO, CSO-CLMDT, PSO-CLMDT, and WSO-CLMDT by 18.52%, 7.77%, 2.27%, 13.14%, 1.67%, 1.07% and 1.04% respectively. The throughput of the PPC-CLMDT model for the round 6 is 0.98, which is higher than the existing CL-CFMPR by 9.47%, WSN-NICC by 4.76%, SMCHE by 3.86%, M-PSO by 7.11%, CSO-CLMDT by 3.74%, PSO-CLMDT by 3.63%, and WSO-CLMDT by 1.81% respectively. Compared to the other existing approaches, the proposed PPC-CLMDT model attain lower computational overhead for 50 nodes under various rounds, which shows the PPC-CLMDT model required minimal amount of processing power to transmit the multimedia data.The improved outcomes of the PPC-CLMDT model are due to its integration of cross-layer optimization over the different layers of the WSN sensor node, which optimize the

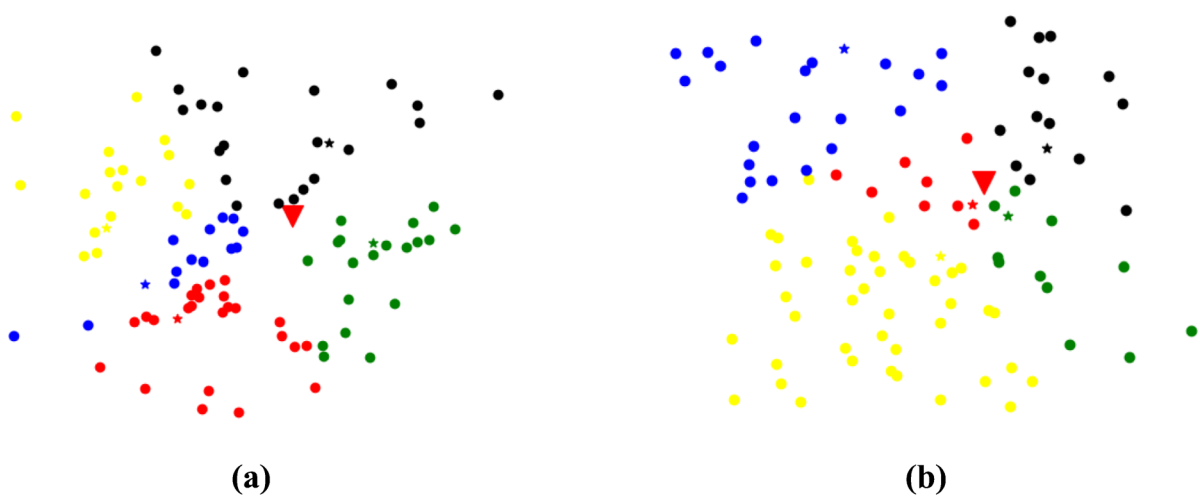


Fig. 4 Simulation Results

Fig. 5 Comparativeanalysis of PPC-CLMDT with 50 nodes

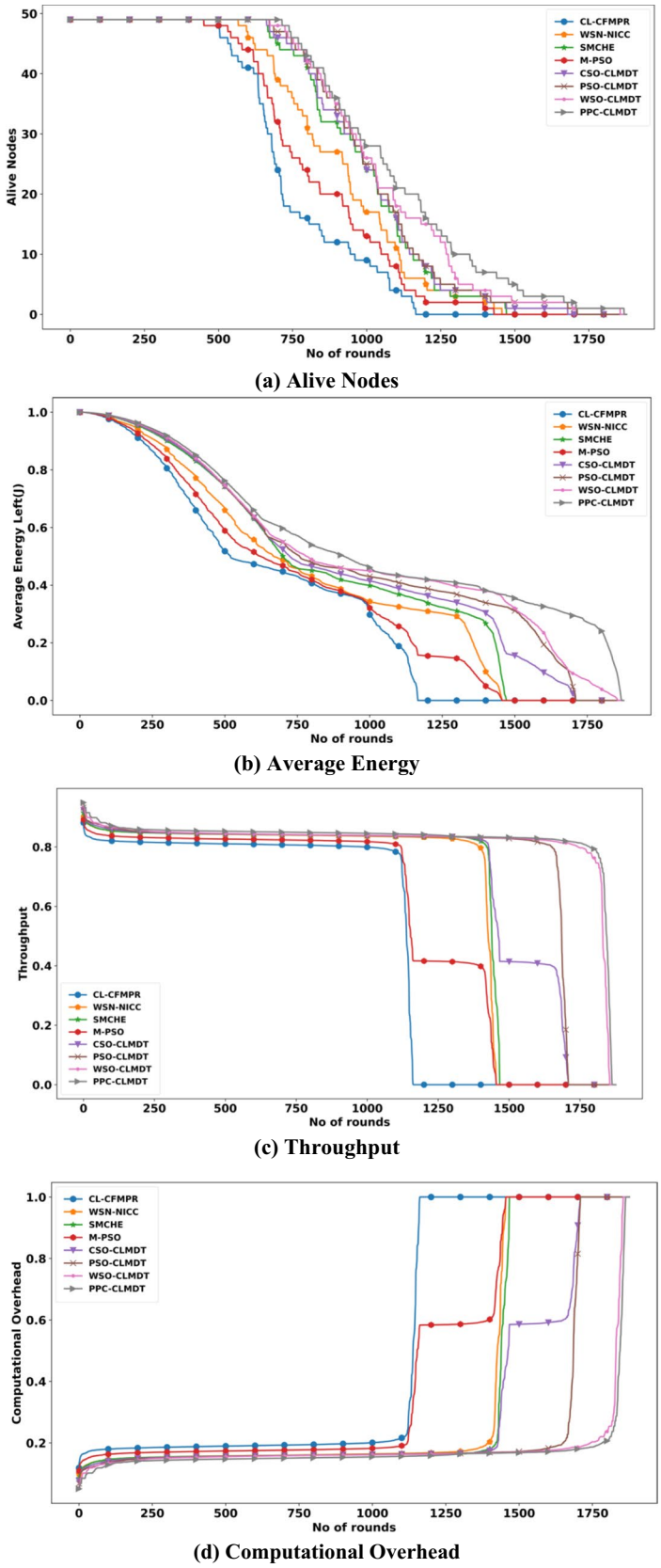


Fig. 6 Comparativeanalysis of PPC-CLMDT with 100 nodes

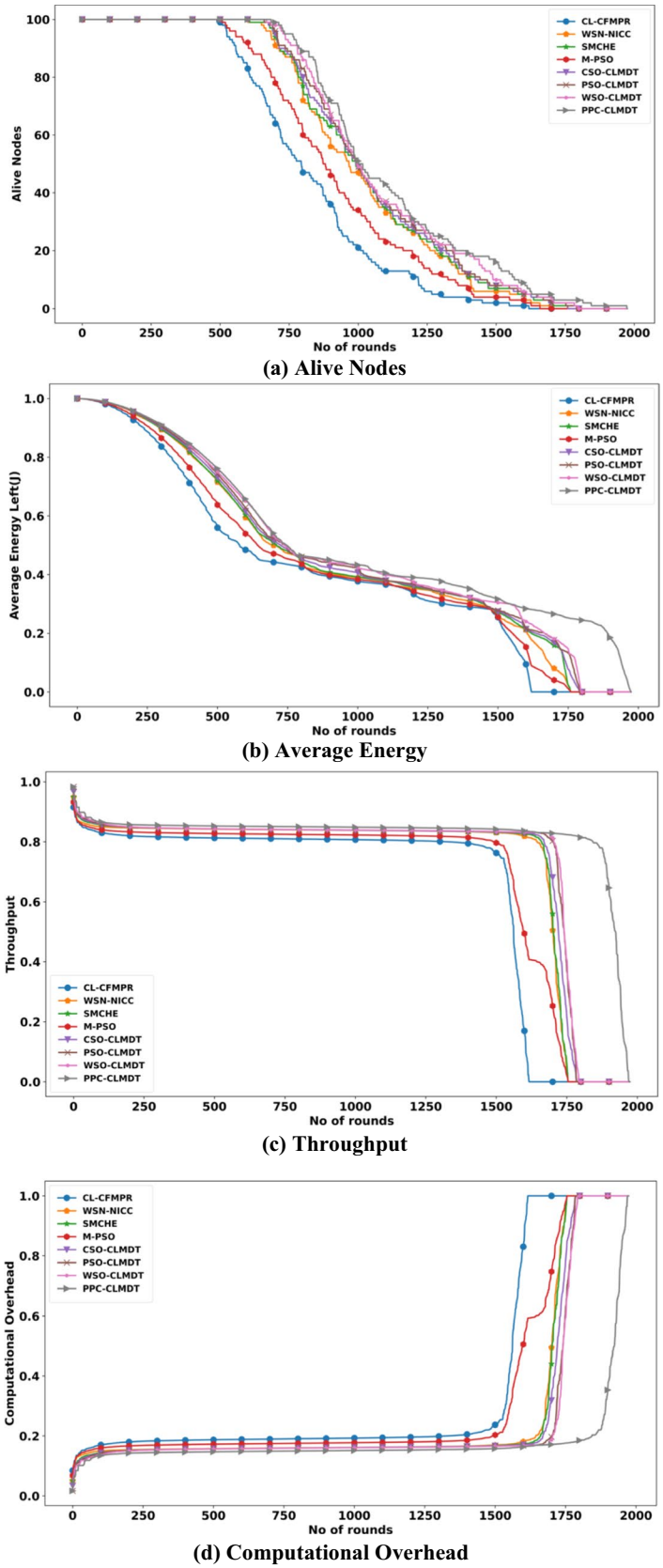


Fig. 7 Comparativeanalysis of PPC-CLMDT with 150 nodes

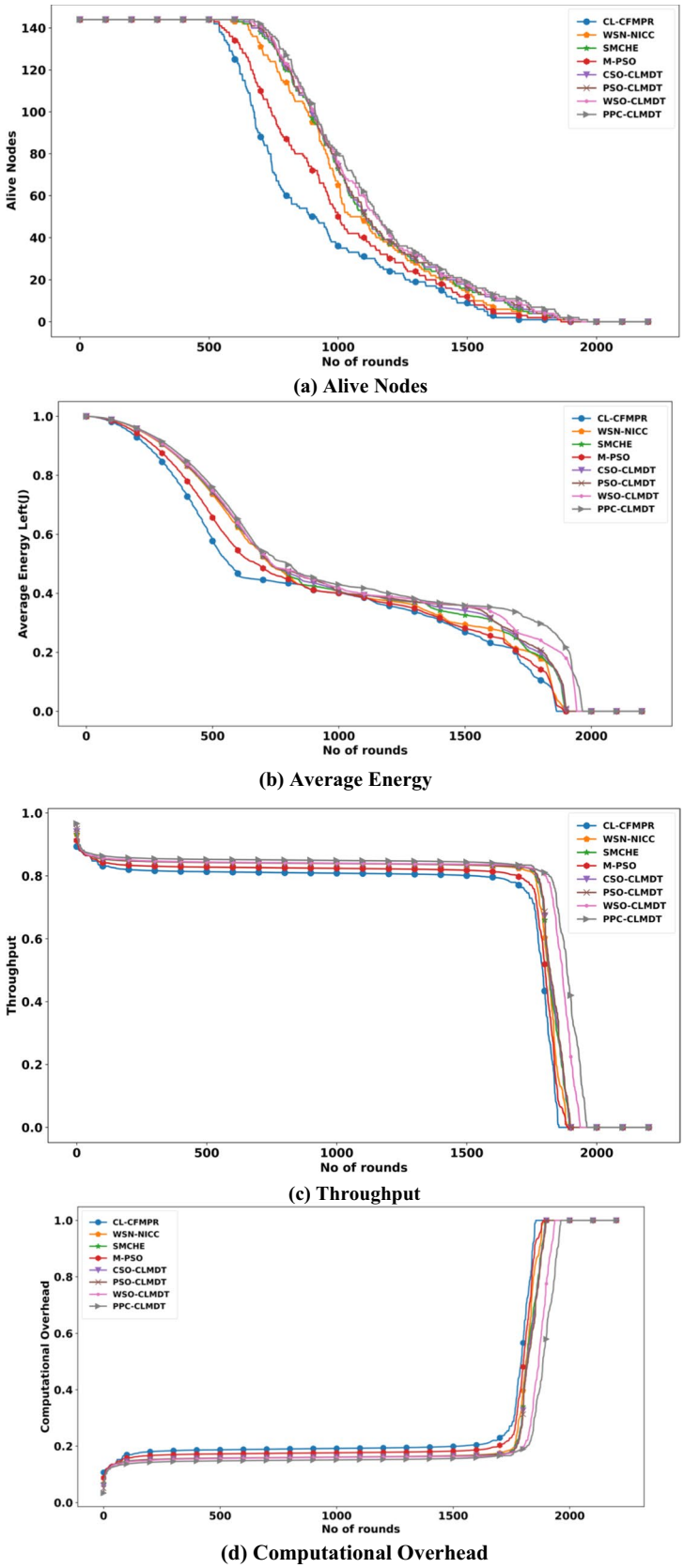


Fig. 8 Comparativeanalysis of PPC-CLMDT with 200 nodes

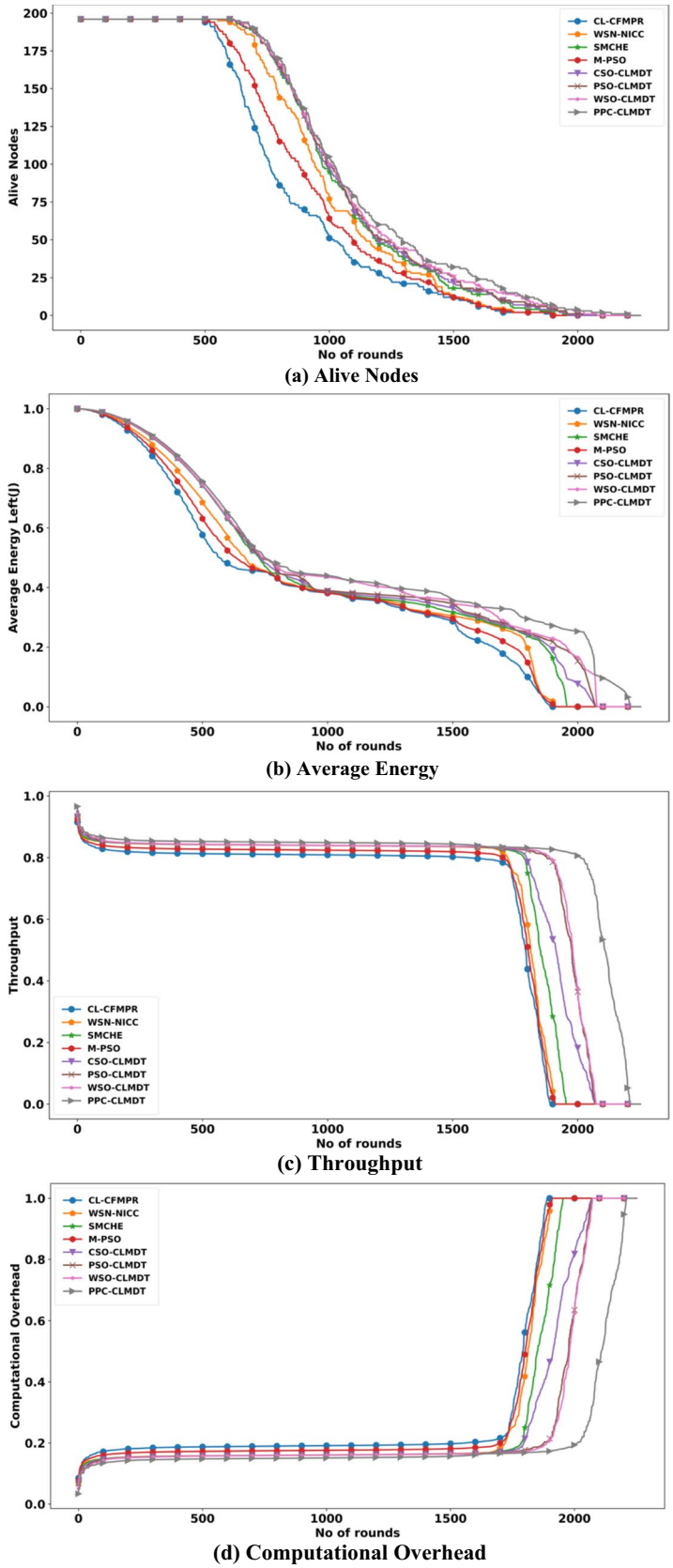
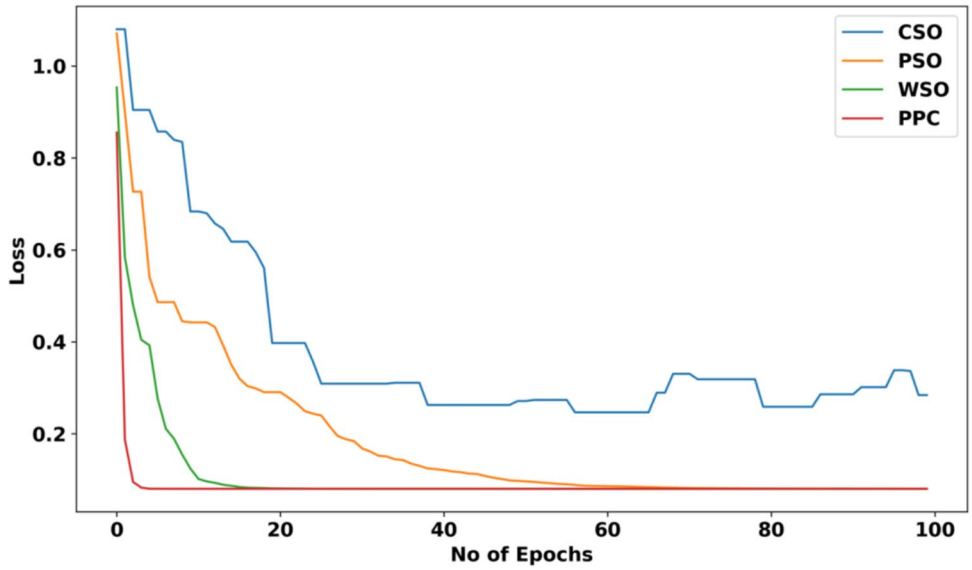


Fig. 9 Convergence analysis



energy consumption and offered better results in multimedia transmission.

b) Comparative Analysis for 100 nodes

The comparative analysis of PPC-CLMDT and existing approaches with 100 nodes is shown in Fig. 6, where the total number of alive nodes for 1000 rounds in the CL-CFMPR model is 21, and the total number of alive nodes for 1000 rounds in the WSN-NICC model is 47, SMCHE is 49, M-PSO is 34, CSO-CLMDT is 49, PSO-CLMDT is 49, and WSO-CLMDT is 49. Finally, the total number of alive nodes for 1000 rounds in the PPC-CLMDT model is 51.The average energy of the PPC-CLMDT model for the round 100 is 0.433 J, which is higher than the existing CL-CFMPR by 12.94%, WSN-NICC by 9.93%, SMCHE by 9.63%, M-PSO by 11.43%, CSO-CLMDT by 6.06%, PSO-CLMDT by 2.49%, and WSO-CLMDT by 2.25% respectively.The throughput of 100 nodes of the PPC-CLMDT method for the 10 rounds is higher than the existing CL-CFMPR, WSN-NICC, SMCHE, M-PSO, CSO-CLMDT, PSO-CLMDT, and WSO-CLMDT by 5.68%, 5.45%, 3.63%, 5.56%, 3.63%, 3.633% and 2.61% respectively.For the round 10, the PPC-CLMDT model attained lower computational overhead of 0.067, which is minimal than the existing approaches and they achieved 0.120 for CL-CFMPR, 0.118 for WSN-NICC,0.101 forSMCHE, 0.119 for M-PSO, 0.1016 for CSO-CLMDT, 0.10 for PSO-CLMDT, and 0.092 for WSO-CLMDT respectively. The PPC-CLMDT model outperforms the other existing approachesbecause of the cross layer optimization, which dynamically allocate the network resources, thus the multimedia packets are transmitted with minimal packet lossand lower latency, that improves the throughput and energy efficiency of the PPC-CLMDT model than the other existing approaches.

c) Comparative Analysis for 150 nodes

The comparative analysis of PPC-CLMDT and existing approaches with 150 nodes is shown in Fig. 7, where the total number of alive nodes for 1000 rounds in the CL-CFMPR model is 36, and the total number of alive nodes for 1000 rounds in the WSN-NICC model is 65, SMCHE is 73, M-PSO is 50, CSO-CLMDT is 74, PSO-CLMDT is 74, and WSO-CLMDT is 76. Finally, the total number of alive nodes for 1000 rounds in the PPC-CLMDT model is 80, it shows the alive nodes of the proposed PPC-CLMDT is greater than the existing approaches. The average energy of 150 nodes of the PPC-CLMDT method for the 400 rounds is higher than the existing CL-CFMPR, WSN-NICC, SMCHE, M-PSO, CSO-CLMDT, PSO-CLMDT, and WSO-CLMDT by 14.25%, 2.14%, 1.88%, 8.19%, 1.88%, 1.87% and 1.22% respectively.The throughput of the PPC-CLMDT model for the round 1200 is 0.84, which is higher than the existing CL-CFMPR by 4.85%, WSN-NICC by 1.159%, SMCHE by 1.154%, M-PSO by 3.007%, CSO-CLMDT by 1.1523%, PSO-CLMDT by 1.150%, and WSO-CLMDT by 1.03% respectively.The proposed PPC-CLMDT model attain lower computational overhead for 50 nodes under various rounds, when it is compared to the other existing approaches, and shows the PPC-CLMDT model required minimal amount of processing power to transmit the multimedia data.

d) Comparative Analysis for 200 nodes

The comparative analysis of PPC-CLMDT and existing approaches with 200 nodes is shown in Fig. 8, where the total number of alive nodes for 1100 rounds in the CL-CFMPR model is 35, and the total number of alive nodes for 1100 rounds in the WSN-NICC model is 62, SMCHE is 66, M-PSO is 48, CSO-CLMDT is 68, PSO-CLMDT is 71, and WSO-CLMDT is 73. Finally, the total number of alive nodes for 1100 rounds in the PPC-CLMDT model is 79. The average energy of the PPC-CLMDT model for the

Table 3 Comparative discussion

Methods	50 Nodes			100 Nodes			150 Nodes			200 Nodes		
	Alive	Average Energy Left	Throughput	Alive	Average Energy Left	Throughput	Alive	Average Energy Left	Throughput	Alive	Average Energy Left	Throughput
CL-CFMPR	9	0.30	0.80	21	0.38	0.81	36	0.40	0.81	51	0.38	0.81
WSN-NICC	17	0.34	0.84	47	0.39	0.84	65	0.40	0.84	77	0.38	0.84
CSO-CLMDT	24	0.40	0.84	49	0.39	0.84	73	0.40	0.84	95	0.39	0.84
PSO-CLMDT	25	0.43	0.84	49	0.42	0.84	74	0.41	0.84	100	0.39	0.84
WSO-LMDT	26	0.45	0.84	49	0.42	0.84	76	0.42	0.84	100	0.44	0.84
PPC-CLMDT	28	0.46	0.85	51	0.43	0.85	80	0.43	0.85	105	0.44	0.85

round 800 is 0.481 J, which is higher than the existing CL-CFMPR by 10.49%, WSN-NICC by 10.10%, SMCHE by 7.28%, M-PSO by 10.300%, CSO-CLMDT by 6.11%, PSO-CLMDT by 4.95%, and WSO-CLMDT by 2.70% respectively. The throughput of 100 nodes of the PPC-CLMDT method for the 150 rounds is higher than the existing CL-CFMPR, WSN-NICC, SMCHE, M-PSO, CSO-CLMDT, PSO-CLMDT, and WSO-CLMDT by 4.42%, 1.43%, 1.37%, 2.93%, 1.27%, 1.18% and 1.125% respectively.The PPC-CLMDT model attained lower computational overhead of 0.050 for the round 4, which is lower than the existing approaches and they achieved 0.110 for CL-CFMPR, 0.101 for WSN-NICC,0.1016 forSMCHE, 0.1061 for M-PSO, 0.1009 for CSO-CLMDT, 0.1002 for PSO-CLMDT, and 0.095 for WSO-CLMDT respectively.Due to the integration of Cross-layer optimization in the PPC-CLMDT model enables intelligent adaptation to the varying network condition, this improves the overall throughput and energy efficiency of the PPC-CLMDT model than the other existing approaches.

7.4 Convergence analysis

The convergence analysis of PPC algorithm is shown in Fig. 9 and it is compared with the existing algorithms, such as CSO, PSO, and WSO. The PPC achieved the lower loss of 0.855, which is smaller than the existing approaches. The existing CSO, PSO, and WSO approaches attains the loss value of 1.08, 1.07, and 0.95 respectively. These comparison displays that the PPC algorithm achieved faster convergence rate, which shows that the PPC-CLMDT method exhibited lower computational complexities.

7.5 Comparative discussion

The advancement of the PPC-CLMDT model is analyzed by a comparative discussion with the already existing model. The data is transmitted with varying numbers of nodes in a loop for nearly 2,000 rounds, and the influence of each method shows the nodes' lifetime improvement for an increasing number of rounds. The suggested approachesare failed to improve QoS, and attained higher communication delay and overhead. Due to the deployment of large scale of sensor nodes, the scalability issues also arised in the existing approaches. These issues are overcomed by the integration of cross-layer optimization approach over sensor layer layers, such as Network layer, MAC layer, and Physical layers. Also, the cross-layer parameter optimization minimizes the transmission delay and raise the overall QoS for each node in the network. The scalability issues are solved through the PPC optimizer by optimizing the WSN's sensor nodes in the cross layer. Table 3 shows the comparison between the existing method and proposed model at 1000 rounds for various nodes.

8 Conclusion

The challenges posed by MMTin WSNs due to their resource-constrained nature have been addressed in this paper through a cross-layer approach. This approach optimizes CH selection and routing of multimedia data without delay, ensuring efficient communication across the heterogeneous sensor nodes comprising WSNs. The proposed methodology leverages PPC optimizer, to enhance network clustering and path selection methods, resulting in robust solutions for multimedia data transmission within WSNs. By implementing PPC, this research significantly advances the capabilities of WSNs in handling multimedia traffic, contributing to the development of more reliable and efficient WSNs that achieve both energy efficiency and minimum data transmission delay. The PPC-CLMDT model is tested in comparison with other methods for varying nodes of 50,100,150, and 200 nodes for around 2000 rounds. The research results demonstrate a reduction in node power utilization and an increase in throughput, affirming the effectiveness of the cross-layer optimization method. Further, the future work could be incorporating the standard machine learning like reinforcement learning or federated learning to improve the routing and dynamic clustering in multimedia WSNs. Moreover, the practical deployment scenarios, such as environmental scenarios and smart healthcare will be explored to contextualize the application of PPC.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Ethical approval Not Applicable.

Informed consent Not Applicable.

Conflict of interest The authors declare no competing interests.

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